Lecture 3. Cholesky Decomposition

Existence and uniqueness, algorithm and cost, pivoting, applications, sweep operator

Contents

3.1 The Cholesky decomposition **2**

3.2 Applications **3**

3.3 The sweep operator **4**

3.4 Selected exercises **5**

Mantra 2

The structure should be exploited whenever possible in solving a problem.

Common structures are symmetry, positive (semi)definiteness, sparsity, Kronecker products and low rank. Statisticians mostly meet positive (semi)definite matrices: covariance matrices and Gram matrices $X^\top X$. A matrix M is positive definite (semidefinite) if $x^\top Mx > 0$ (≥ 0) for all $x \neq 0$; since $x^\top Mx = x^\top \frac{M+M^\top}{2} x$, we may assume M symmetric.

3.1 The Cholesky decomposition

Motivation. Statistical computations constantly solve systems $A\mathbf{x} = \mathbf{b}$ with A symmetric positive definite (normal equations, Gaussian likelihoods). Cholesky factors such A as $LL^\top$ at half the cost of LU, with no pivoting and excellent stability.

Theorem 3.1 (Cholesky)

If $A \in \mathbb{R}^{n\times n}$ is symmetric positive definite, then $A = LL^\top$ for a unique lower triangular L with positive diagonal entries.

Proof. Induction on n. For $n = 1$, $a = \sqrt{a}\sqrt{a}$ with $a > 0$. For $n > 1$, partition and match blocks:

$$\begin{pmatrix} a_{11} & \mathbf{a}^\top \\ \mathbf{a} & A_{22} \end{pmatrix} = \begin{pmatrix} \ell_{11} & \mathbf{0}^\top \\ \mathbf{b} & L_{22} \end{pmatrix} \begin{pmatrix} \ell_{11} & \mathbf{b}^\top \\ \mathbf{0} & L_{22}^\top \end{pmatrix} \iff a_{11} = \ell_{11}^2, \quad \mathbf{a} = \ell_{11}\mathbf{b}, \quad A_{22} = \mathbf{b}\mathbf{b}^\top + L_{22}L_{22}^\top.$$

Since $a_{11} = e_1^\top Ae_1 > 0$, the conditions force $\ell_{11} = \sqrt{a_{11}}$ and $\mathbf{b} = \mathbf{a}/\sqrt{a_{11}}$. (Every principal submatrix of a positive definite matrix is positive definite, e.g. $\mathbf{y}^\top A_{22}\mathbf{y} = (0, \mathbf{y}^\top)A(0, \mathbf{y}^\top)^\top > 0$; but the recursion needs more.) The **Schur complement** $S = A_{22} - \mathbf{b}\mathbf{b}^\top = A_{22} - \mathbf{a}\mathbf{a}^\top/a_{11}$ is positive definite: for $\mathbf{y} \neq \mathbf{0}$, put $x = (-\mathbf{a}^\top\mathbf{y}/a_{11}, \mathbf{y}^\top)^\top \neq 0$; then

$$0 < x^\top Ax = \frac{(\mathbf{a}^\top\mathbf{y})^2}{a_{11}} - 2\frac{(\mathbf{a}^\top\mathbf{y})^2}{a_{11}} + \mathbf{y}^\top A_{22}\mathbf{y} = \mathbf{y}^\top S\mathbf{y}.$$

By the induction hypothesis $S = L_{22}L_{22}^\top$ for a unique such L_{22}, which completes existence and uniqueness. ■

The proof is the algorithm: compute the first column of L, update the Schur complement, and recurse; this can be done in place in the lower triangle of A.

Cost. The Schur complement updates are symmetric, so only half the entries are computed:

$$\tfrac{1}{2}\{2(n-1)^2 + 2(n-2)^2 + \cdots + 2\} \approx \tfrac{1}{3}n^3 \text{ flops}$$

plus n square roots, *half* the cost of LU. The algorithm is very stable without pivoting, and it fails (a nonpositive number under a square root) exactly when A is not positive definite. So Cholesky is the cheapest way to *test* positive definiteness.

Example 3.1

For

$$A = \begin{pmatrix} 4 & 6 & 4 & 4 \\ 6 & 37/4 & 11/2 & 6 \\ 4 & 11/2 & 29/4 & 7 \\ 4 & 6 & 7 & 33/4 \end{pmatrix}$$

the first column of L is $(4,6,4,4)^\top/2 = (2,3,2,2)^\top$. The Schur complement $A_{22} - \mathbf{b}\mathbf{b}^\top$ has first column $(37/4 - 9,\ 11/2 - 6,\ 6 - 6)^\top = (1/4, -1/2, 0)^\top$, so $\ell_{22} = 1/2$ and the second column of L is $(1/2, -1, 0)^\top$. Continuing, the pivots are $4, \frac{1}{4}, \frac{9}{4}, \frac{1}{4}$, all positive, and

$$L = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 3 & 1/2 & 0 & 0 \\ 2 & -1 & 3/2 & 0 \\ 2 & 0 & 2 & 1/2 \end{pmatrix}.$$

Since the algorithm ran to completion with positive pivots, $A = LL^\top$ is positive definite (and $\det A = (2 \cdot \frac{1}{2} \cdot \frac{3}{2} \cdot \frac{1}{2})^2 = \frac{9}{16}$).

Pivoting for semidefinite matrices. If A is only positive semidefinite (e.g. $X^\top X$ with X rank deficient), a zero pivot can occur. With **symmetric pivoting**, at stage k the largest remaining diagonal entry is moved (by a symmetric row and column swap) to the pivot position. If the largest remaining diagonal entry is 0, the whole remaining block is 0: a positive semidefinite matrix with zero diagonal entry a_{ii} has zero row i, since $|a_{ij}| \le \sqrt{a_{ii}a_{jj}}$. The algorithm then stops, giving

$$PAP^\top = LL^\top, \qquad L \in \mathbb{R}^{n \times r}, \quad r = \operatorname{rank} A.$$

LAPACK provides `potrf` (no pivoting) and `pstrf` (pivoting); R's `chol` returns the upper factor $R = L^\top$, $A = R^\top R$.

3.2 Applications

Multivariate normal log-likelihood. For $y \sim N_n(0, \Sigma)$,

$$\log f(y) = -\frac{n}{2}\log(2\pi) - \frac{1}{2}\log\det\Sigma - \frac{1}{2}y^\top\Sigma^{-1}y.$$

- Method 1: invert Σ ($2n^3$), form the quadratic form ($2n^2$), compute the determinant by LU ($\frac{2}{3}n^3$).
- Method 2: factor $\Sigma = LL^\top$ ($\frac{1}{3}n^3$), solve $Lx = y$ by forward substitution (n^2), then $y^\top\Sigma^{-1}y = x^\top x$ ($2n$) and $\log\det\Sigma = 2\sum_i \log \ell_{ii}$ (n).

Method 2 is about 8 times cheaper and more accurate, and the log-determinant computed this way cannot overflow. To evaluate the density at many points, factor once and pay n^2 per point.

Linear regression. For $X \in \mathbb{R}^{n \times p}$, solve the normal equations $X^\top X\beta = X^\top y$ by Cholesky:

form $X^\top X$	np^2 (symmetric)
form $X^\top y$	$2np$
Cholesky of $X^\top X$	$\frac{1}{3}p^3$
two triangular solves	$2p^2$
$(X^\top X)^{-1}$ for standard errors	$\frac{4}{3}p^3$

In total about $np^2 + \frac{1}{3}p^3$ flops, or $np^2 + \frac{5}{3}p^3$ with standard errors. This is the fastest method when $n \gg p$, but forming $X^\top X$ squares the condition number (Lecture 4).

3.3 The sweep operator

For a symmetric $A = (a_{ij}) \in \mathbb{R}^{n\times n}$ with $a_{kk} \neq 0$, **sweeping** on the kth diagonal entry maps A to $\hat{A} = (\hat{a}_{ij})$ with

$$\hat{a}_{kk} = -\frac{1}{a_{kk}}, \qquad \hat{a}_{ik} = \frac{a_{ik}}{a_{kk}}, \qquad \hat{a}_{kj} = \frac{a_{kj}}{a_{kk}}, \qquad \hat{a}_{ij} = a_{ij} - \frac{a_{ik}a_{kj}}{a_{kk}} \quad (i, j \neq k).$$

The last formula is one step of Gaussian elimination (a Schur complement update), so a sweep costs about n^2 flops ($n^2/2$ using symmetry), and it can be done in place. The **inverse sweep** on entry k is

$$\check{a}_{kk} = -\frac{1}{a_{kk}}, \qquad \check{a}_{ik} = -\frac{a_{ik}}{a_{kk}}, \qquad \check{a}_{kj} = -\frac{a_{kj}}{a_{kk}}, \qquad \check{a}_{ij} = a_{ij} - \frac{a_{ik}a_{kj}}{a_{kk}};$$

substituting $\hat{A}$ shows directly that it restores A (e.g. $\hat{a}_{ij} - \hat{a}_{ik}\hat{a}_{kj}/\hat{a}_{kk} = a_{ij} - a_{ik}a_{kj}/a_{kk} + a_{ik}a_{kj}/a_{kk} = a_{ij}$).

Lemma 3.2 (Exchange property)

Let $U = [u_1, \ldots, u_n]$ and $V = [v_1, \ldots, v_n]$ be $p \times n$ with $V = UA$. After sweeping on entry k,

$$[v_1, \ldots, v_{k-1}, -u_k, v_{k+1}, \ldots, v_n] = [u_1, \ldots, u_{k-1}, v_k, u_{k+1}, \ldots, u_n]\,\hat{A}.$$

That is, sweeping exchanges the roles of u_k and v_k (up to sign).

Proof. Column k of the right side is $-v_k/a_{kk} + \sum_{i\neq k} u_i a_{ik}/a_{kk} = -(v_k - \sum_{i\neq k} u_i a_{ik})/a_{kk} = -u_k a_{kk}/a_{kk} = -u_k$, using $v_k = \sum_i u_i a_{ik}$. For $j \neq k$, column j is

$$v_k \frac{a_{kj}}{a_{kk}} + \sum_{i\neq k} u_i \Big(a_{ij} - \frac{a_{ik}a_{kj}}{a_{kk}}\Big) = \sum_{i\neq k} u_i a_{ij} + \frac{a_{kj}}{a_{kk}}\Big(v_k - \sum_{i\neq k} u_i a_{ik}\Big) = \sum_{i\neq k} u_i a_{ij} + a_{kj}u_k = v_j.$$

■

Proposition 3.3

Partition $A = \left(\begin{smallmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{smallmatrix}\right)$ with $A_{11} \in \mathbb{R}^{k\times k}$ invertible. Sweeping on the diagonal entries $1, \ldots, k$ (in any order, as long as each pivot is nonzero) gives

$$\hat{A} = \begin{pmatrix} -A_{11}^{-1} & A_{11}^{-1}A_{12} \\ A_{21}A_{11}^{-1} & A_{22} - A_{21}A_{11}^{-1}A_{12} \end{pmatrix}.$$

Proof. Apply the lemma to $A = IA$ ($U = I$, $V = A$) for each sweep. After sweeping $1, \ldots, k$,

$$\begin{pmatrix} -I_k & A_{12} \\ 0 & A_{22} \end{pmatrix} = \begin{pmatrix} A_{11} & 0 \\ A_{21} & I \end{pmatrix} \begin{pmatrix} \hat{A}_{11} & \hat{A}_{12} \\ \hat{A}_{21} & \hat{A}_{22} \end{pmatrix}.$$

Reading off blocks: $A_{11}\hat{A}_{11} = -I$, $A_{11}\hat{A}_{12} = A_{12}$, $A_{21}\hat{A}_{11} + \hat{A}_{21} = 0$ and $A_{21}\hat{A}_{12} + \hat{A}_{22} = A_{22}$, which give the four blocks. The result depends only on the set of swept indices, so sweeps commute. ■

Regression by sweeping. Sweep the $(p+1)\times(p+1)$ matrix $\left(\begin{smallmatrix} X^\top X & X^\top y \\ y^\top X & y^\top y \end{smallmatrix}\right)$ on its first p diagonal entries. By the proposition the result is

$$\begin{pmatrix} -(X^\top X)^{-1} & (X^\top X)^{-1}X^\top y \\ y^\top X(X^\top X)^{-1} & y^\top y - y^\top X(X^\top X)^{-1}X^\top y \end{pmatrix} = \begin{pmatrix} -\sigma^{-2}\operatorname{Cov}(\hat\beta) & \hat\beta \\ \hat\beta^\top & \|y-\hat y\|^2 \end{pmatrix}:$$

the coefficients, their (unscaled) covariance and the residual sum of squares appear at once. This makes sweeping ideal for **stepwise regression**. Adding variable j to the current model is one sweep on entry j and removing it is one inverse sweep, each $O(p^2)$, instead of refitting from scratch; the swept matrix always holds the current $\hat\beta$, standard errors and RSS, from which the F (or $t^2=\hat\beta_j^2/\widehat{\operatorname{Var}}\hat\beta_j$) statistics for entering or leaving variables are read off. A pivot that is (numerically) zero signals a variable that is collinear with those already in the model. The price is that it works with $X^\top X$, with the conditioning of the normal equations.

3.4 Selected exercises

Motivation. The first exercise connects Cholesky with LU and gives a determinant test for definiteness; the second proves the entry bound behind symmetric pivoting; the third shows what one sweep does to the residual sum of squares in stepwise regression.

Exercise 3.1 ($LDL^\top$ and a determinant test)

Let $A\in\mathbb{R}^{n\times n}$ be symmetric with nonsingular leading principal submatrices $A_k=A[1{:}k,1{:}k]$, $k=1,\dots,n$, and put $\det A_0=1$.
(a) Show that $A=LDL^\top$ for a unique unit lower triangular L and diagonal D, and that $d_k=\det A_k/\det A_{k-1}$.
(b) Show that A is positive definite if and only if $\det A_k>0$ for all k, and that then $LD^{1/2}$ is the Cholesky factor of A.

Solution. [Supplement]

(a) *Existence.* $A=A_n$ is nonsingular, so by Theorem 2.2 $A=LU$ uniquely with L unit lower triangular. As in the remark after that theorem, $A_k=L_kU_k$ for the leading blocks, so $\det A_k=\prod_{i\le k}u_{ii}$; hence $u_{kk}=\det A_k/\det A_{k-1}\neq 0$. Let $D=\operatorname{diag}(u_{11},\dots,u_{nn})$ and $M=D^{-1}U$, which is unit upper triangular. Then $A=LDM$ and, by symmetry, $A=A^\top=M^\top(DL^\top)$. Here $M^\top$ is unit lower triangular and $DL^\top$ is upper triangular, so this is another LU decomposition of A; by uniqueness $M^\top=L$, i.e. $A=LDL^\top$.

Uniqueness. If $A=L'D'L'^\top$ with L' unit lower triangular and D' diagonal, then $L'(D'L'^\top)$ is an LU decomposition, so $L'=L$ and $D'L^\top=DL^\top$; comparing diagonals, $D'=D$.

(b) If A is positive definite, so is each A_k (take x supported on the first k coordinates), and $\det A_k$ is the product of its eigenvalues, which are positive. Conversely, if all $\det A_k>0$, then by (a) all $d_k>0$, and for $x\neq 0$, with $y=L^\top x\neq 0$ (L is invertible),

$$x^\top Ax=y^\top Dy=\sum_k d_k y_k^2>0.$$

Then $A=(LD^{1/2})(LD^{1/2})^\top$ with $LD^{1/2}$ lower triangular with positive diagonal $\sqrt{d_k}$, which by the uniqueness in Theorem 3.1 is the Cholesky factor. In Example 3.1, the pivots $4,\frac14,\frac94,\frac14$ are exactly the d_k, and $\ell_{kk}=\sqrt{d_k}$. ■

Exercise 3.2 (Entries of a positive semidefinite matrix)

Let A be symmetric positive semidefinite. Show:
(a) $|a_{ij}| \le \sqrt{a_{ii}a_{jj}}$ for all i, j;
(b) if $a_{ii} = 0$, then the whole i-th row and column of A vanish;
(c) $\max_{i,j}|a_{ij}| = \max_i a_{ii}$.

Solution. [Supplement] For real s, t and $i \neq j$, positive semidefiniteness gives

$$0 \le (se_i + te_j)^\top A(se_i + te_j) = s^2 a_{ii} + 2st\, a_{ij} + t^2 a_{jj}. \tag{$\dagger$}$$

(b) If $a_{ii} = 0$, take $t = 1$: $2sa_{ij} + a_{jj} \ge 0$ for all $s \in \mathbb{R}$. A linear function of s bounded below is constant, so $a_{ij} = 0$ for every $j \neq i$; by symmetry the column vanishes too.

(a) If $a_{ii} = 0$ or $a_{jj} = 0$, both sides are 0 by (b). If $a_{ii} > 0$, take $s = -a_{ij}$, $t = a_{ii}$ in ($\dagger$): $0 \le a_{ij}^2 a_{ii} - 2a_{ij}^2 a_{ii} + a_{ii}^2 a_{jj} = a_{ii}(a_{ii}a_{jj} - a_{ij}^2)$, so $a_{ij}^2 \le a_{ii}a_{jj}$. The case $i = j$ is trivial.

(c) By (a), $|a_{ij}| \le \sqrt{a_{ii}a_{jj}} \le \max\{a_{ii}, a_{jj}\}$, and the diagonal entries themselves are among the $|a_{ij}|$.

This justifies symmetric pivoting for semidefinite Cholesky: the largest remaining diagonal entry dominates the whole remaining block, and if it is 0 the block is 0. ■

Exercise 3.3 (What one sweep does in stepwise regression)

Let $X_1 \in \mathbb{R}^{n\times q}$ and $x \in \mathbb{R}^n$ be such that $[X_1, x]$ has full column rank, $y \in \mathbb{R}^n$, $H_1 = X_1(X_1^\top X_1)^{-1}X_1^\top$, $r_x = (I - H_1)x$ and $r_y = (I - H_1)y$. Order the cross-product matrix of (X_1, x, y) as

$$T = \begin{pmatrix} X_1^\top X_1 & X_1^\top x & X_1^\top y \\ x^\top X_1 & x^\top x & x^\top y \\ y^\top X_1 & y^\top x & y^\top y \end{pmatrix}.$$

(a) Show that after sweeping on the first q diagonal entries, the lower-right 2×2 block is $\begin{pmatrix} \|r_x\|^2 & r_x^\top r_y \\ r_y^\top r_x & \|r_y\|^2 \end{pmatrix}$. In particular the next pivot is nonzero.
(b) Sweeping next on the entry of x, show that the residual sum of squares drops from $\|r_y\|^2$ to $\|r_y\|^2 - (r_x^\top r_y)^2/\|r_x\|^2$, and that the coefficient of x in the regression of y on $[X_1, x]$ is $r_x^\top r_y/\|r_x\|^2$.

Solution. [Supplement]

(a) Apply Proposition 3.3 with $A_{11} = X_1^\top X_1$ (invertible since X_1 has full column rank) and $A_{21} = [x, y]^\top X_1$, $A_{22} = [x, y]^\top [x, y]$. The swept lower-right block is

$$A_{22} - A_{21}A_{11}^{-1}A_{12} = [x, y]^\top (I - H_1)[x, y].$$

Since $I - H_1$ is symmetric and idempotent, $u^\top(I - H_1)v = \{(I - H_1)u\}^\top(I - H_1)v$, which gives the four entries. The pivot $\|r_x\|^2$ is positive because $x \notin \mathcal{R}(X_1)$ (full column rank of $[X_1, x]$), so $r_x \neq 0$.

(b) One sweep on the x-entry, applied to the current entries $a_{xx} = \|r_x\|^2$, $a_{xy} = r_x^\top r_y$, $a_{yy} = \|r_y\|^2$, gives by definition

$$\hat{a}_{yy} = a_{yy} - \frac{a_{yx}a_{xy}}{a_{xx}} = \|r_y\|^2 - \frac{(r_x^\top r_y)^2}{\|r_x\|^2}, \qquad \hat{a}_{xy} = \frac{a_{xy}}{a_{xx}} = \frac{r_x^\top r_y}{\|r_x\|^2}.$$

By Proposition 3.3 the result of the $q+1$ sweeps depends only on the swept set, so it equals T swept on all entries of $X = [X_1, x]$. By the regression-by-sweeping identity, its (y, y) entry is $\|y - \hat{y}\|^2$ for the regression on X and its (x, y) entry is the last coordinate of $\hat{\beta} = (X^\top X)^{-1}X^\top y$, the coefficient of x. The drop $(r_x^\top r_y)^2/\|r_x\|^2$ is nonnegative and vanishes iff $r_x \perp r_y$; the coefficient is the slope of the residuals r_y on r_x (the Frisch–Waugh theorem). ■